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Math 104C

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Numerical Solutions to the Nonlinear Drug Diffusion Problem

This problem studied the drug diffusion problem modeled by a nonlinear boundary value equation and describes how the concentration of a drug diffuses through tissue. The three numerical methods used to approximate the solution were the linear finite difference, nonlinear shooting combined with Newton's method, and the nonlinear finite differences with Newton's method. After using python to generate the different plots, I was able to see that the linear finite difference method produced a smooth approximation for the diffusion model. The nonlinear shooting and nonlinear finite difference methods were then used to solve the nonlinear part of the problem. Newton's method was applied to obtain the convergence. Numerical tests showed that both of the nonlinear approaches converged quickly, with the error decreasing rapidly after only a few iterations. The nonlinear finite difference method had a slightly faster convergence and had more stability, while the shooting method also gave highly accurate results. The experiment overall confirmed that Newton based nonlinear methods are effective for solving nonlinear drug diffusion models.

Introduction & Mathematical Background

Drug diffusion models are used to describe how a drug moves through a tissue and how it is absorbed or consumed. In this project, the concentration of the drug is affected by the position and the nonlinear absorption. The nonlinear term tells us that the absorption rate does not increase proportionally when the concentration of the drug becomes too large . Boundary conditions are added in so that the concentration is fixed within the left boundary and a zero in the right boundary. This represents the diffusion in a finite region.

The Mathematical model for drug concentration C(x) is given by:

$$DC''(x) \; - \; \frac{\{V_{\{max\}}C(x)\}}{\{K_m + C(x)\}} = 0$$

Where D is the diffusion coefficient, Vmax is the maximum absorption rate and Km is the saturation constant (kubota). After nondimensionalization the equation we get is:

$$u''(x) \; = \; \lambda \frac{u(x)}{\kappa + u(x)} \; , \; u(0) \; = \; 1 \; , \; u(1) \; = \; 0$$

Here λ measures the strength of diffusion and absorption. κ is the dimensionless saturation parameter. In order to simplify this model, the assumption C <<Km is used. Under this, the denominator is approximated by Km, which leads us to the following equation:

$$u''(x) \; = \; \frac{\lambda}{\kappa} u(x), \; u(0) = 1, \; u(1) = 0$$

Three numerical methods were used to solve these problems. The linearized equation was solved using the linear finite difference method, the nonlinear problem was solved using a nonlinear shooting method combined with Newton's method, and with a nonlinear finite difference method using Newton's method to solve the nonlinear algebraic system.The finite difference method and Newton's method used in this project are standard techniques in numerical analysis (Burden and Faires) . These methods were then compared in terms of convergence behavior, accuracy, and stability.

Implementation and Setup

I implemented the problem into visual studio using python. Using the base parameters λ = 2 and κ = .3 . For my plots, I used N = 80 interior grid points. Other values such as 20,40,80 and 160 can be used to compare convergence.

For the linear finite difference method, the interval [0,1] was divided into evenly spaced grid points. The second derivative was then approximated using centered finite differences, which then gave us a tridiagonal linear system. The boundary conditions used were u(0) = 1 and u(1) = 0 on the right hand side and then solved using numpy.linalg.solve

For the nonlinear shooting method, the unknown initial slope of s = u'(0) was given the initial guess of -1. Newton's method was used with residual $\phi(s) = u(1; s)$ which measures the mismatch from the boundary condition u(1) = 0 . The IVP was then advanced using a fourth order Runge-Kutta method.

For the nonlinear finite differences, the initial guess was a straight line from u(0)= 1 to u(1)=0. At each of the newton steps, a jacobian system was formed and solved using numpy.linalg.solve. Newton iterations kept going until the residual was below 10^-8, with a maximum of 100 iterations. The code used was numpy, matplotlib, and numpy.linalg.solve

Results and Visualizations

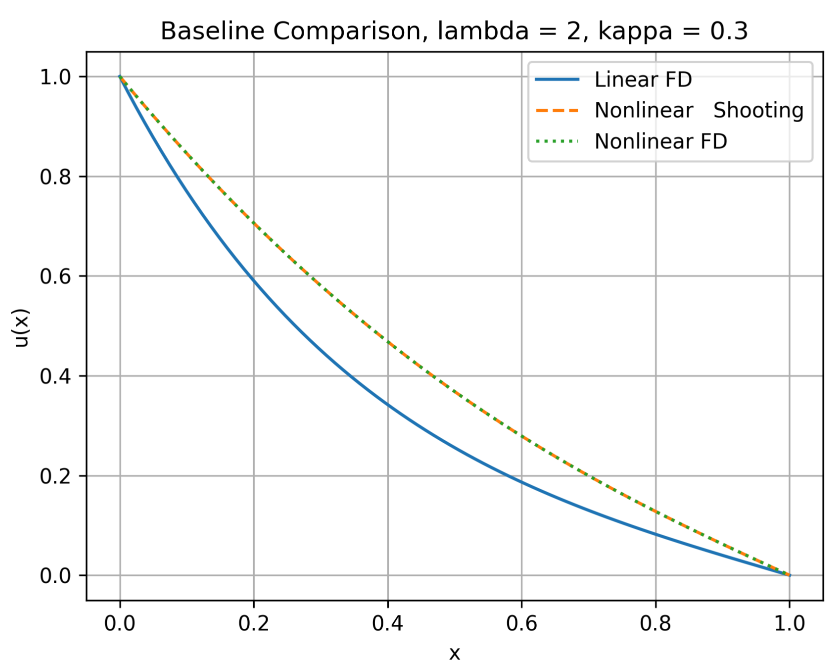

Figure 1. Comparison of the linear and nonlinear numerical solutions for $\lambda = 2$ and κ=0.3

The methods were tested using the parameters $\lambda = 2$ and $\kappa = .3$. Figure one above compares the linear finite difference solution, the nonlinear shooting solution, and the nonlinear finite difference solution. All three methods produced smooth concentration profiles that satisfied the boundary conditions u(0) = 1 and u(1) = 0. The nonlinear shooting method and the nonlinear finite difference method solutions were very similar, almost being identical, showing a strong agreement between them. The linearized finite difference solution also had a similar trend but was slightly different due to the linearization assumption of C << Km.

N	u(.25)	u(.50)	u(.75)	J_0
20	.64197785	.36844623	.16293293	1.58254588
40	.64179031	.36825496	.16285732	1.60013687
80	.64174056	.36820440	.16283774	1.60932851
160	.64172774	.36819139	.16283276	1.61402847

Figure 2. Grid refinement results showing u(0.25), u(0.50),, u(0.75),, and the incoming flux J0

A grid refinement study was also done using the values of N = 20,40,80,160. In figure 2 we have the values at u(.25), u(.50), u(.75) and J_0 = -u'(0). The values only changed slightly as the grid was refined, showing stability and numerical convergence in the nonlinear finite difference method.

The nonlinear shooting method and the nonlinear finite difference method produced almost identical solutions for $\lambda = 2$ and $\kappa = .3$.The maximum difference between the two solutions was approximately 3.75 x 10^-6, indicating great agreement. Newton's method converged in 5 iterations while the nonlinear finite difference method converged in only 4 iterations. The quick convergence shows us how fast Newton's corrections decrease towards being precise. This also shows us that both approaches are accurate and efficient in solving the drug diffusion boundary problem.

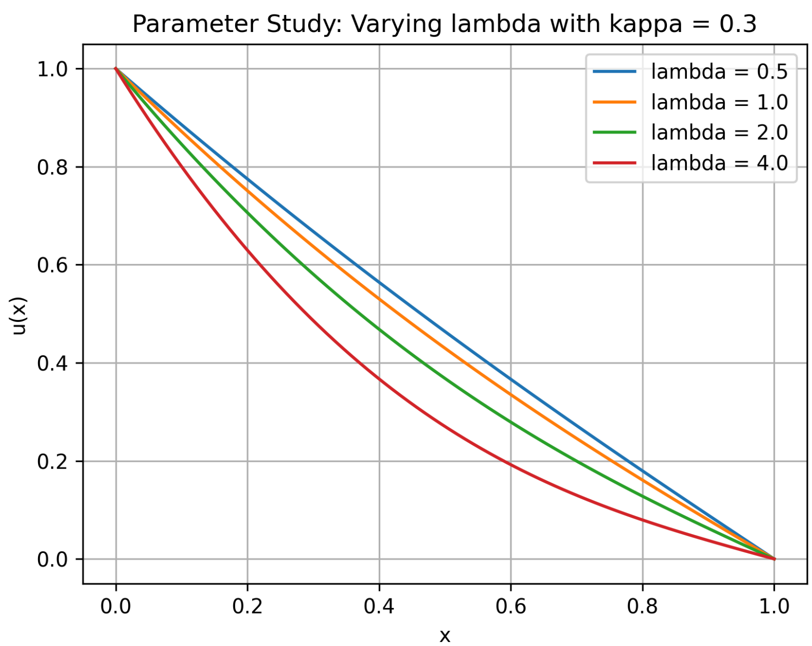

Figure 3. Nonlinear concentration profiles for different values of λ with κ=0.3

In order to study the effect of the parameter λ , I ran the nonlinear finite difference method for values λ =.5,1,2 and 4, while keeping κ as .3. The computations were implemented in python using the NumPy library for numerical calculations and Matplotlib for visualization purposes . The results in figure 3 showed that changing λ affected the shape of the solution, as λ increases the curve bends more, showing a stronger nonlinear absorption effect. Smaller values of λ gave a more gradual decrease from u(0) = 1 and u(1) = 0.

Discussion and Analysis

The linearized model produced a solution that followed a similar decreasing trend as the nonlinear model, however there were some small noticeable differences. The linear model assumes that the concentration remains smaller than the saturation parameter, given by $u \ll \kappa$. Using this assumption, the nonlinear absorption term can be approximated by a linear expression. For the base case $\kappa = .3$, the concentration is comparable to or larger than κ over the domain. Due to this the linearized finite difference solution underestimates the concentration compared to the nonlinear solutions. This causes the linearized model to be only accurate when the concentration remains small relative to κ .

The nonlinear shooting method and the nonlinear finite difference method produced very similar results. The maximum difference between the two numerical solutions was approximately 3.75 x 10 ^-6. The shooting method was slightly more difficult to use because it required determining the unknown initial slope u'(0). Also the nonlinear finite difference method only needed 4 iterations as opposed to 5 from the shooting method, making it more efficient and it also enforced the boundary conditions.

The parameter showed that increasing λ significantly changed the concentration profile. As λ went from .5 to 4 , the solution became strongly curved and decreased at a faster rate across the domain. This corresponds to stronger absorption relative to diffusion, causing the drug concentration to drop quicker as it moves through the tissue . The incoming flux $J_0 = -u'(0)$ also increased as the solution became steeper near x = 0. For the base case of $\lambda = 2$ the grid refinement study showed J_0 to be approximately 1.61, indicating that a substantial influx of drug is required to maintain the concentration during absorption.

One limitation of the computation is the initial guess in Newton's method. The shooting method required an initial slope guess of $s_0 = -1$, and choosing a bad value could potentially slow down convergence or make more complex problems tricky to solve. Another limitation could be grid resolution. In this project the grid refinement showed that the flux J_0 changed from 1.5725 for N = 20 to 1.6140 for N = 160. While luckily the grid showed that the values would converge, finer grids could require larger linear systems and more effort to compute. Also, Newton's method relies on solving linear systems accurately at each iteration which would be challenging for larger and more strongly nonlinear problems.

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